Why Catastrophe Insurance is Different

- Catastrophe losses are increasing worldwide. This isn't just a Canadian problem.
- Governments around the world are facing the same challenge of financing increasingly costly disasters.

Drivers of Increasing Catastrophe Losses

1. **Hazard**
   - Increasing frequency and severity of secondary perils
   - Climate change alters the frequency, severity, and spatial distribution of weather-related perils.
   - Leads to more intense storms, prolonged wildfire seasons, and heavier precipitation events
   - Also leads to compounding events such as storm plus flooding or wildfire plus wind
2. **Vulnerability**
   - The replacement cost of structures, infrastructure, and technology escalates with time and as real wealth grows.
   - Most regions have inadequate flood and wildfire mitigation strategies.
   - With the rise of technology, new risk features like solar panels and batteries have been created.
3. **Exposure**
   - Population growth in high-risk areas creates denser pockets of value, turning what would have been a localized natural event into a multi-billion-dollar catastrophe.
   - It also means that there are simply more high-value physical assets sitting in harm's way.
4. **Socio-economic effects**
   - Post-disaster surges in demand for labour and raw materials, combined with general economic inflation, significantly push up the repair and rebuilding cost.

Why catastrophe risks challenge private insurance markets?

1. **Losses are highly correlated**
   - A single event can generate thousands of claims simultaneously.
   - Risk cannot be diversified within a region.
2. **Loss distributions are fat-tailed**
   - Rare events produce extremely large losses.

- Extreme losses dominate total losses.

3. **Extreme losses require large capital**
   - Insurers must hold capital to survive rare but severe events.
   - Capital tied to low-probability risks is often expensive.
4. **Tail risk can threaten insurer solvency**
   - Tail risk refers to the possibility of extremely rare but exceptionally large losses.
   - Extreme catastrophe losses may exceed available industry capital.
   - Reinsurance capacity may be limited during large events.

Designing Catastrophe Insurance Programs

Catastrophe Insurance Programme Policy Goals

- **Affordability:** Insurance premiums remain within reach for households and businesses, including those in higher-risk areas.
- **Availability:** Coverage is widely offered so that most properties can obtain catastrophe insurance.
- **Equity/Fairness:** The distribution of costs and benefits across policyholders and taxpayers is viewed as fair.
- **Resilience:** this includes both economic recovery and the continued functioning and stability of insurance markets.
- **Reducing the protection gap:** A larger share of catastrophe losses is covered by insurance rather than borne by households or governments. Basically you want to minimize uninsured losses. **Protection Gap = Total Loss - Insured Loss**
- **Solvency:** insurers and the system remain able to pay claims after very large disasters.
- **Sustainability:** the program remains financially and politically viable over time.

How does the nature of the parallel shape see a program design?

- How the insurance industry typically treats the peril?
  - Included in standard property policy
  - Covered through an endorsement
  - Or typically excluded. This usually involves government intervention to establish availability.
- Potential losses relative to available financial capacity of
  - The domestic economy
  - The primary insurance market

    - Global Reinsurance Capacity
- Return period of peak peril
    - How frequently catastrophic events occur?
    - How quickly must capital be replenished?
- Insurability Trends
    - Climate trends may undermine long-term insurability.

Six categories that structure CI programs

1. Coverage Scope, and Participation

- This defines the exact boundary lines of the insurance program.
- The peril scope could be single peril, which focuses on one specific hazard, or multi peril, which bundles multiple natural hazards together.
- Programs decide whether to protect only residential homes or include commercial properties and public infrastructure. Expanding property types increases the total exposure but broadens the premium base.
- **Coverage structure:** The policy could be either automatically bundled inside standard home and property policies or sold separately as an optional endorsement or add-on.
- Participation Rules dictate who is required or permitted to operate within a CII program across three distinct levels:
    1. **Insurer participation**
        - If mandatory, all insurers writing standard property coverage in the market must cede risk to or participate in the public catastrophe pool. Ensures broad risk sharing and **avoids adverse selection.**
        - If voluntary, insurers choose whether to cede risk or use the program.
    2. **Coverage offer mandate**
        - If mandatory, primary insurers are legally forced to offer catastrophe coverage to policyholders.
        - If voluntary, insurers can choose whether or not to make catastrophic coverage available.
    3. **Policy Holder Purchase Mandate**
        - If mandatory property owners are legally required to buy the coverage.
        - If voluntary property owners decide whether to purchase the endorsement or the policy.

2. Pricing & Cross Subsidization

- The key decision is to choose between the risk-based pricing versus cross-subsidization.
- **Risk Based Pricing**

    - Under risk-based pricing, your premium reflects your exact expected loss plus reinsurance and regulatory capital costs.
    - A beachfront home or a house in a dense forest interface pays an extremely high premium, while a house on high ground or away from trees pays a low premium.
    - It sends clear financial warnings to property owners and deters people from building in high-risk flood plains or wildfire zones and awards mitigation.
    - It also ensures that the pool actually collects enough money to cover expected losses.
    - A downside is that premiums in extreme hazard zones can become very expensive, forcing people to drop the coverage, widening the protection gap.
- **Cross-subsidization**
    - Under cross-subsidization, low-risk policyholders pay more than their actual risk to help cover the costs of high-risk policyholders.
    - Governments use four main levers to cross-subsidize:
        1. **Flat rate premiums:** Everyone pays the exact same flat rate or percentage, regardless of location.
        2. **Implicit cross-subsidies:** Averaging rates regionally or nationally, so lower-risk zones pad the fund for higher-risk zones
        3. **Capped Premiums:** Capping maximum rates in high-risk zones so homeowners aren't priced out
        4. **Explicit government subsidies:** Direct taxpayer cash injected into the pool to lower costs for policyholders
    - The core problem with cross-subsidies is that they create moral hazard. If living on a dangerous flood plain is heavily subsidized by urban condo owners, people have zero financial incentive to move out or reinforce their homes.

3. Capitalization & Funding

- Initial Program Capitalization describes how the Catastrophe Insurance Program builds its starting balance sheet on day one.
    - **Government Capital or Disaster Funds:** Direct government grants, seed loans, or public disaster funds provide immediate capacity.
    - **Insurer contributions:** Primary insurance companies are required to contribute initial capital or equity to set up the pool.
    - **No initial fund:** The program starts with $0 in reserves and relies entirely on post-disaster borrowing, taxation, or industry assessments if an event occurs.
- Program funding discusses how the program continuously finances losses across decades. There are three main funding approaches:
    1. **Pre-funded**
        - Premiums are collected from policyholders continuously and accumulate in our dedicated catastrophe reserve fund before a disaster strikes.

- Money is immediately available when claims roll in, which improves national financial preparedness.
- Requires higher initial premiums and large accumulated pools of public money, which face political pressure to be diverted for other public spending over time.

2. **Post Event Funded**
   - Funds are raised only after a disaster occurs via emergency government borrowing, post-event taxes or special assessments on insurers.
   - Keeps upfront consumer premiums lower.
   - Creates immense fiscal risk for governments and delays claims payments while funding is arranged.
3. **Hybrid**
   - A blend of accumulated reserves, private reinsurance, and post-event backstops or borrowing authority.
   - Balances upfront premium affordability with post-disaster financial resilience.

4. Pooling & Governance

- Market structure and governance outline: who manages the pool and how decisions are made.
- The primary trade-off is between public/state control and private/market governance.
  - **State-owned:**
    - The Government creates, owns, and directly manages the Catastrophe Insurance Pool.
    - This leads to strong public accountability, a direct government backstop, and focuses on policyholder protection rather than profit.
    - The major con is that it is vulnerable to political interference.
  - **Private Mutual or Insurer Consortium:**
    - Leverages private insurance expertise and removes political pressure from pricing decisions.
    - A con is that it may prioritize insurer solvency over broader public affordability goals.
- Governance dictates how the board is structured and who holds voting power over critical decisions.
  - **Board Composition:**
    - Boards dominated by politicians tend to freeze rates, whereas boards dominated by insurers may raise rates aggressively, which widens the protection gap.
    - The key decision is to choose whether the board consists of elected public officials, independent industry experts, insurance executives, or

consumer advocates.

- **Operational Distribution:**
  - In almost all successful models, private primary insurers handle sales, policy administration, and claims processing. The government rarely builds its own consumer-facing claims department.
  - The central entity manages the aggregated reserve pool, buys global reinsurance or issues CAT bonds.

- Another important decision is to choose whether catastrophe insurance policies will be designed nationally or regionally.
  - If it's a **national framework**, it has a common coverage, eligibility, and program rules across the country, which ensures a consistent consumer experience.
  - If it's a **regional or decentralized framework**, rules vary across jurisdictions or insurers. Coverage and pricing reflect local conditions and regulatory choices, which provide greater flexibility but less consistency.

5. Government's Role

- The government's rules are often the outcome of earlier design choices.
- Here are some of the roles that the government can play in a catastrophe insurance program.
  - **Government as insurer**
    - Assumes the primary insurance obligation
    - May retain or delegate pricing, underwriting, policy servicing, and claims management
    - This is the greatest direct government involvement.
  - **Government is re-insurer**
    - Accepts catastrophe risk transferred from primary insurers.
    - Expands market capacity and supports insurer solvency.
    - Bears losses within the reinsurance layer
  - **Government as regulator**
    - Establishes program rules, coverage requirements, and insurer obligations.
    - Also sets solvency standards and may influence market behavior.
  - **Government as backstop**
    - Provides protection only for extreme losses above a specific threshold.
    - Supports insurers or reinsurers against catastrophic tail risk and participates only in the most extreme catastrophe losses.
  - **Government as Liquidity Provider**
    - Provides immediate financing following catastrophic events.
    - Usually in the form of a loan, and the funding is expected to be recovered through future assessments or repayments.

- Provides immediate claims-paying capacity without necessarily creating permanent public losses.

6. Operational Roles

- The core question is to decide who performs key insurance functions in the program.
- Most policies rely on private insurers for selling and servicing.
- **Pricing:**
  - Pricing can be set by a public entity or by private insurers.
  - Public pricing can improve affordability but may weaken risk-based signals.
- **Underwriting:**
  - Risks can be underwritten by a public entity or by private insurers, or through a shared arrangement.
  - Public underwriting can expand coverage but may crowd out private market participation.
- **Claims Management:**
  - Again, claims can be handled by a public entity or by private insurers or by a shared arrangement.
  - Shared claims administration can use existing infrastructure but may lead to coordination problems and coverage disputes.

Canada's Current Approach

- Canada manages catastrophe risk through a layered system rather than a national catastrophe insurance program.
- Households retain losses below deductibles, above limits, and uninsured losses. Insurance may be unavailable or limited for some catastrophe risks.
- Private insurers provide residential property coverage and bear primary underwriting risk.
- Reinsurers absorb large correlated catastrophe losses. Pricing is influenced by global catastrophe losses.
- The Federal Disaster Financial Assistance Arrangements (DFAA) Programme provides financial assistance to provinces after major disasters. It's more of a backstop rather than an insurance program.

Canadian catastrophe events in 2025

Peril	Provinces Affected	Number of Events
Flood	BC, AB, SK, MC, ON, QC, NB, NS, NL	6

Peril	Provinces Affected	Number of Events
Wildfire	AB, SK, MB, NL	5
Severe convective storm	AB, QC, SK, MB	3
Winter weather	BC, AB, ON, QC, NB, NS, NL	3
Windstorm	ON, QC	1

- Most losses in Canada are driven by secondary perils.
- Insured catastrophe losses in Canada have increased significantly over the past two decades.
- Several recent events have produced multi-billion dollar insured losses, such as severe convective storms, floodings, and wildfires.

Earthquake Risk in Canada

- Earthquakes are low-frequency but potentially catastrophic events that can threaten the solvency of insurers.
- Earthquake risk is concentrated in British Columbia and Eastern Quebec.
- Earthquake insurance is purchased as an optional endorsement to homeowners' policies.
- Take-up rates vary significantly across regions.
  - In Quebec, there is a very low take-up rate and limited awareness, which contributes to a large protection gap.
  - In British Columbia, there is a high take-up rate, which creates a highly concentrated insurance exposure.
- Even insured households may face significant uninsured losses.
- Designing earthquake insurance requires balancing tail risk management, affordability, and protection gap reduction.

Flood Risk in Canada

- Flooding is the most frequent and costly natural disaster in Canada.
- Flood risk arises from river flooding, urban storm water flooding, and coastal flooding.
- Overland residential flood insurance was introduced in 2015 but remains difficult to obtain for many high-risk properties.
- Climate change and continued development in flood-prone areas are increasing both the frequency of losses and the value of exposed property.
- Therefore, many high-risk households remain uninsured or underinsured, creating a significant protection gap.
- Designing flood insurance requires balancing affordability, availability, risk-based pricing, and protection gap reduction.

Wildfire Risk in Canada

- Wildfires have become more frequent and severe in recent years.
- Risk is concentrated in wildland-urban interfaces where development meets forested land.
- Wildfire losses are covered under standard property insurance policies.
- Large correlated events can generate significant accumulation risk for insurers, which is why some insurers have begun restricting new business or increasing premiums in high-risk areas.
- Wildfire is a natural and ecologically important process, creating policy trade-offs around suppression, mitigation, and development.
- Designing wildfire insurance requires balancing insurance availability, affordability, and incentives for risk reduction.